

THEME 5: INTEGRATION

Unit 5.1 Antiderivatives

Source: Stewart, Section 4.9 pp 356 - 364.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of an antiderivative of a function;
2. be able to find the most general antiderivative of a function;
3. be able to find a function if the derivative(s) of the function is(are) given.

Remarks

1. You can omit the section about linear motion (pp 360 - 361).
2. Memorize the table on p 358.

Assigned Problems

Stewart, Exercises 4.9 p 361: All the odd numbers from 5 to 25, 31, 39, 45, 47, 55, 65, 67.

Unit 5.2 The area problem

Source: Stewart, Section 5.1 pp 372 - 384.

Learning objectives

On completion of this unit you should

1. be able to estimate the area of a region that lies between the graph of a continuous function, the x -axis and the lines $x = a$ and $x = b$ by using the sum of the areas of rectangles. The rectangles are constructed by using as sample points either the left endpoints or the right endpoints of each subinterval;
2. be able to find an upper and lower estimate of the area of a region that lies between the graph of a continuous function, the x -axis and the lines $x = a$ and $x = b$. The function must be either increasing or decreasing on the interval.

Remark: Omit Example 2

Assigned Problems

Stewart, Exercises 5.1 p 381: Numbers 1, 4, 7, 9.

Unit 5.3 The definite integral

Source: Stewart, Section 5.2 pp 384, 385, 391 and 393.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of a definite integral;
2. know what the integrand and limits of integration of a definite integral are;
3. be able to use a Riemann sum to approximate a definite integral;
4. know and be able to use the properties of a definite integral;
5. be able to evaluate integrals by interpreting it in terms of areas;
6. know the comparison properties of the integral.

Remarks

1. Study only Examples 1,5 and 8.
2. If f is integrable on an interval $[a, b]$ and $f(x) \geq 0$ for every $x \in [a, b]$, then $\int_a^b f(x)dx \geq 0$. In fact, $\int_a^b f(x)dx$ is equal to the area of the region between the graph of f , the x -axis and the lines $x = a$ and $x = b$.
3. If f is integrable on an interval $[a, b]$ and $f(x) \leq 0$ for every $x \in [a, b]$, then $\int_a^b f(x)dx \leq 0$. In fact, $\int_a^b f(x)dx$ is equal to the negative of the area of the region between the graph of f , the x -axis and the lines $x = a$ and $x = b$.

Assigned Problems

Stewart, Exercises 5.2 p 394: Numbers 5(a&b), 7, 35, 41, 43, 45, 59, 60, 67.

Unit 5.4 The Fundamental Theorem of Calculus

Source: Stewart, Section 5.3 pp 402 - 409.

Learning objectives

On completion of this unit you should know and be able to proof and use the Fundamental Theorem of Calculus, part 2 (p 403).

Remarks

1. The following proof will be discussed in class:

Theorem 1. *The Fundamental Theorem of Calculus.*

If f is continuous on $[a, b]$, then $\int_a^b f(x)dx = F(b) - F(a)$, where F is any antiderivative of f , that is, a function such that $F' = f$.

Proof. Divide $[a, b]$ into n subintervals of equal length $\Delta x = \frac{b-a}{n}$. Let $a = x_0 < x_1 < \dots < x_{n-1} < x_n = b$ be the endpoints of these intervals and let F be an antiderivative of f , that is, $F' = f$. This means

- F is differentiable on $[a, b]$;
- F is continuous on $[a, b]$ (from Theorem 4 on p 157).

We will now apply the Mean Value Theorem to each of the subintervals $[x_{i-1}, x_i]$, $i = 1, 2, \dots, n$. Since

- F is continuous on $[x_{i-1}, x_i]$;
- F is differentiable on (x_{i-1}, x_i) ,

there exists a number c_i in the interval (x_{i-1}, x_i) such that

$$\frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} = F'(c_i) = f(c_i) \quad (\text{it is given that } F' = f). \quad (1)$$

Let this number c_i be the sample point in each interval. As f is continuous on $[a, b]$, f is integrable on $[a, b]$ (Theorem 3, p 386) and

$$\begin{aligned} \int_a^b f(x)dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x, \text{ definition of definite integral (p.384)} \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} \Delta x, \text{ from (1)} \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} (x_i - x_{i-1}), \text{ as } \Delta x \text{ is the length of each subinterval} \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n F(x_i) - F(x_{i-1}) \\ &= \lim_{n \rightarrow \infty} (F(x_1) - F(x_0)) + (F(x_2) - F(x_1)) + \dots + (F(x_n) - F(x_{n-1})) \\ &= \lim_{n \rightarrow \infty} (F(x_n) - F(x_0)) \\ &= F(b) - F(a). \end{aligned}$$

□

2. Work through Examples 5 to 9.

Assigned Problems

Stewart, Exercises 5.3 p 406: 21, 22, All the odd numbers between 25 and 55, 63.

Unit 5.5 Indefinite integrals

Source: Stewart, Section 5.4 pp 409 - 412.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of an indefinite integral;
2. be able to evaluate indefinite integrals.

Assigned Problems

1. Stewart, Exercises 5.4 p 415: Number 1, all the odd numbers from 7 to 21, 39, 42, 48, 53.
 2. Stewart, Exercises 5.5 p 425: All the odd numbers from 1 to 53 and from 59 to 73.
Integrate without making a substitution.
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THEME 6: VECTOR ALGEBRA

Unit 6.1 Three-dimensional coordinate systems

Source: Stewart, Section 12.1 pp 830 - 836.

Learning objectives

On completion of this unit you should

1. be able to explain what a three-dimensional rectangular coordinate system is;
2. be able to calculate the distance between two points in the three-dimensional space;
3. be able to find the projections of a given point on coordinate planes or coordinate axes;
4. be able to write down the equation of a sphere with given centre and radius.

Assigned Problems

Stewart, Exercises 12.1 p 835: Numbers 3, 4, 9, 10, 13, 15, 17.

Unit 6.2 Vectors

Source: Stewart, Section 12.2 pp 836 - 846.

Learning objectives

On completion of this unit you should

1. know what a vector is;